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1 Future discussions

More Kernel Optimizations for SGEMM

We have only scratched the surface of optimizing SGEMM. Other topics include:

1. **Memory coalescing:** arrange accesses so neighboring threads use neighboring addresses and require fewer memory transactions.
2. **Occupancy and register pressure:** balance per-thread resources against the number of warps available to hide latency.
3. **Shared-memory bank conflicts:** pad or rearrange tiles so threads in a warp do not serialize their accesses.
4. **Vectorized memory operations:** use aligned `float4` loads and stores to reduce instruction overhead.
5. **Warp tiling:** divide a block's output tile among warps to improve locality and register reuse.
6. **Autotuning:** search tile sizes and launch parameters for the target GPU instead of relying on one universal configuration.

Alternative Programming Models and DSLs

In production, writing raw CUDA is rarely worth the effort.

1. Triton: tiling is at the heart of GPU optimization, so that becomes our new atomic unit instead of threads. Thread-level optimizations are optimized by a JIT compiler.
2. CUTLASS, CuTe DSL, Layout Algebra: Manually handling indexing is cumbersome and error-prone. We can compose common patterns with an algebra that can be checked at compile time.

Kernel Fusion and Specialized Kernels

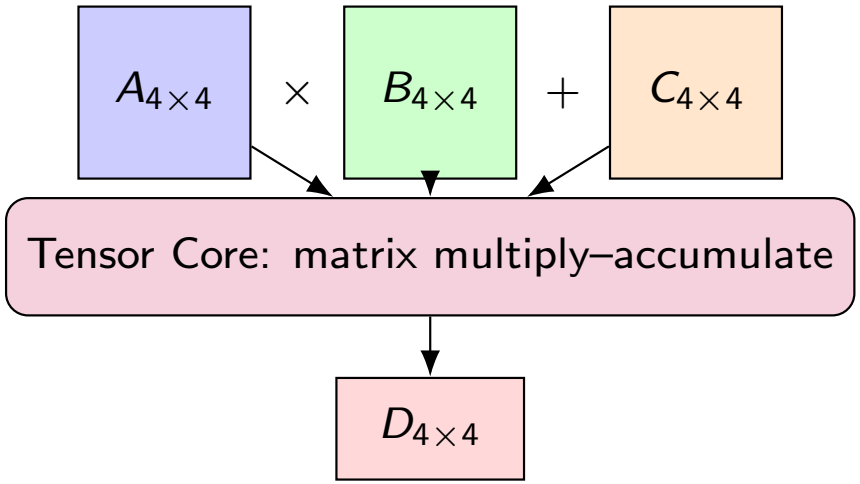
Kernels, be definition, have to read/write from DRAM. We can **fuse** kernels to increase performance.

1. Linear Layer + Nonlinearity = Fused MLP layer
2. FlashAttention series: complex fusion of the Attention Mechanism
3. Megakernels: taken to the extreme, we can fuse the operations necessary for the entire forward pass of a model into a **megakernel.**

Continuing GPU evolution

2017: Volta Tensor Cores

- Tensor Cores perform mixed-precision matrix multiply–accumulate: $D = AB + C$.
- The original Volta unit multiplies FP16 inputs and accumulates in FP16 or FP32.
- Matrix multiplication becomes an explicit hardware primitive, alongside general-purpose CUDA cores.



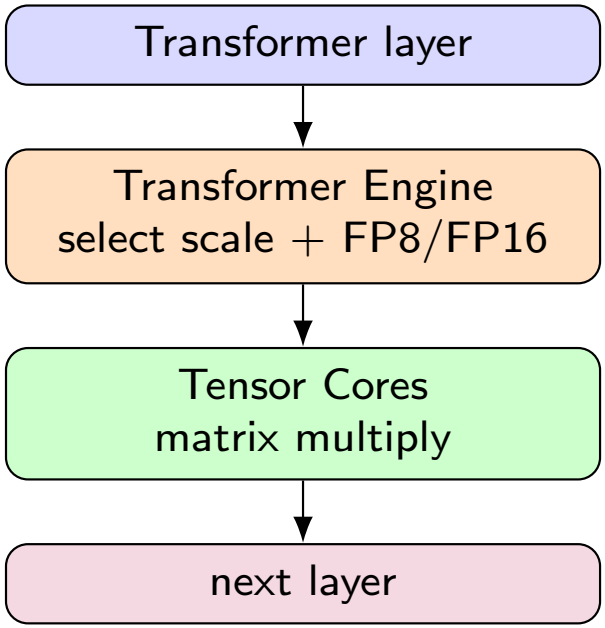
Operation shape shown for first-generation Volta Tensor Cores. Source:

NVIDIA, *Volta Architecture Whitepaper*, 2017.

Continuing GPU evolution

2020 onward: specialize around AI dataflow

- Ampere (2020) adds TF32 and structured sparsity support.
- Hopper (2022) adds FP8 Tensor Cores and a Transformer Engine that dynamically chooses FP8 or FP16 per layer.
- Blackwell (2024) adds more low-precision formats and a second-generation Transformer Engine. This is more than incremental scaling.



Source: NVIDIA, *Hopper Architecture In-Depth* (2022).

Continuing GPU evolution

2026: Vera Rubin, optimizing the rack

- The target is no longer an isolated GPU, but a complete rack-scale computing platform.
- Vera CPUs, Rubin GPUs, memory, NVLink, and networking are designed together to run one distributed AI workload.
- A high-bandwidth scale-up fabric lets many accelerators share model parameters and intermediate results.
- Performance now depends on communication, power, cooling, and software orchestration as well as matrix throughput.

Unit of optimization

Kernel

↓

GPU

↓

Server

↓

Rack-scale system

Source: NVIDIA, *Vera Rubin Architecture*, announced roadmap.

Alternative Accelerator Architectures

Big Picture

With respect to AI accelerators, GPUs are actually remarkably **general**. We can further specialize into AI-specific compute operations.

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3. NPUs: Serving AI workloads in power and heat constrained environments (mobile, consumer electronics, edge devices, etc.).

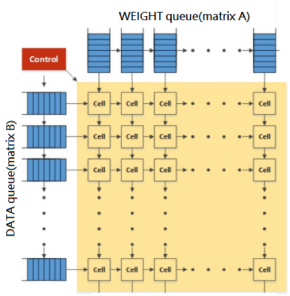
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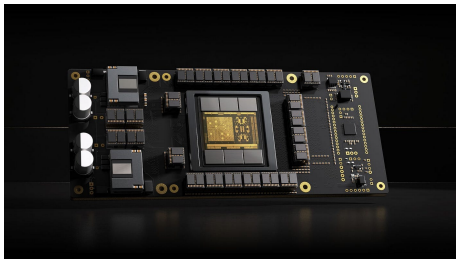
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3. NPUs: Serving AI workloads in power and heat constrained environments (mobile, consumer electronics, edge devices, etc.).
4. WSE (Wafer Scale Engine): Massive data bandwidth through multi-die wafers.

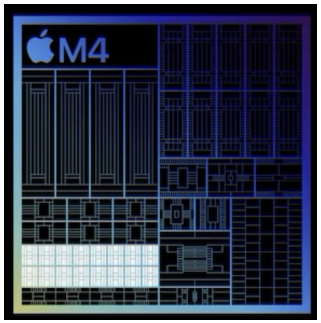
Takeaways



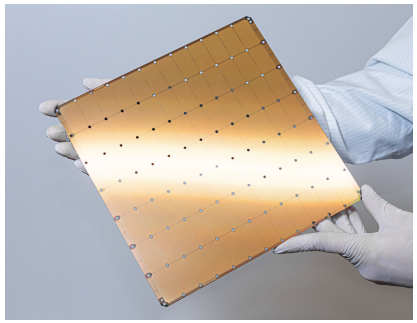
TPU systolic array



Etched Sohu



Apple M4 NPU



Cerebras WSE

There is no universally best architecture

Everything is a tradeoff. Optimizing for something always comes with a cost. **Workflow demands shape hardware evolution.**